GEOMETRIC TRANSFORMATION

Chapter 7 (Intro to GIS by Chang)

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Geometric Transformation

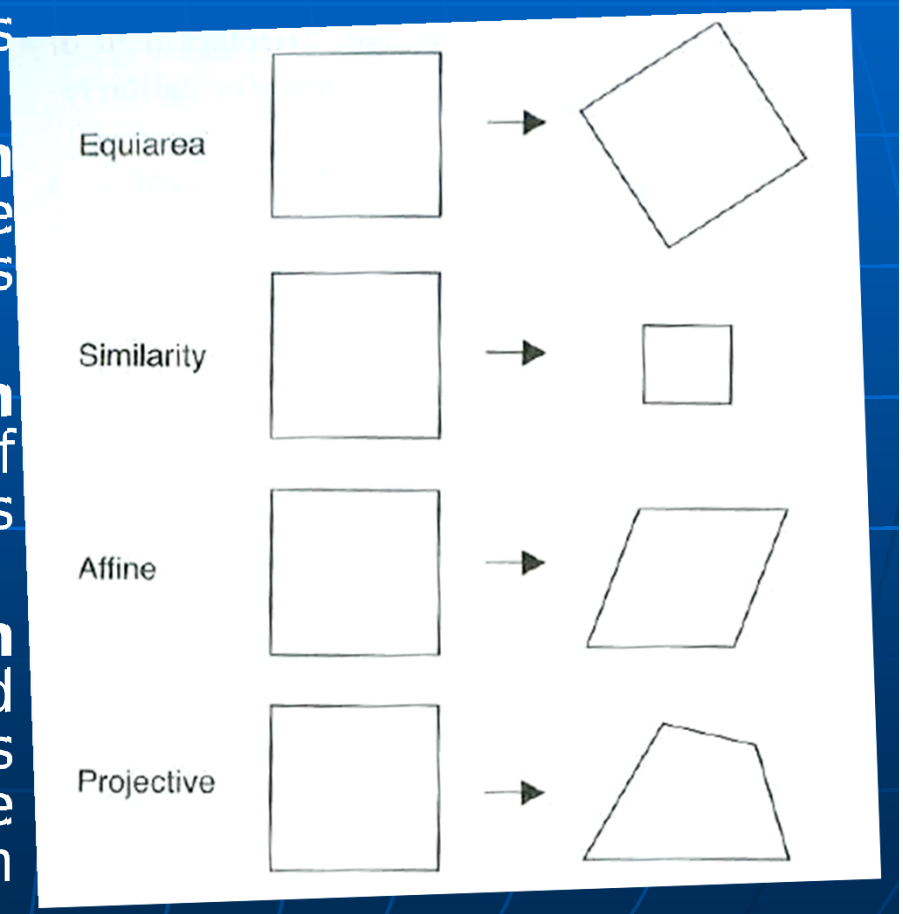
- Geometric transformation is the process of using a set of control points and transformation equations to register a digitized map, a satellite image, or an aerial photograph onto a projected coordinate system

Map to map and Image to map transformation

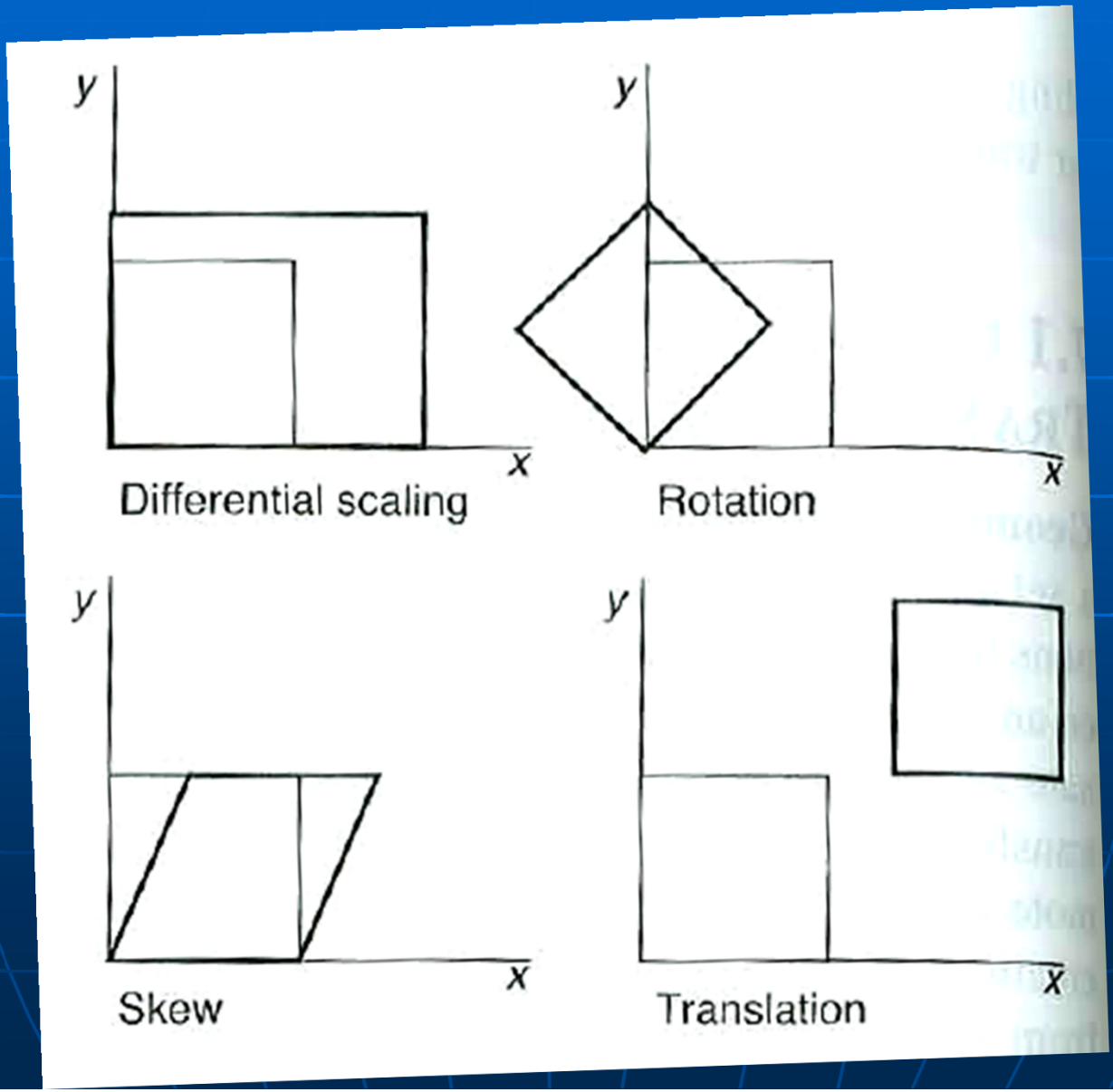
- Geometric transformation converts the newly digitized map into projected coordinates in a process often called **map to map transformation**.
- **Image to map transformation** applies to remotely sensed data.

Transformation methods

- **Equiarea transformation** allows rotation of the rectangle and preserves its shape and size.
- **Similarity transformation** allows rotation of the rectangle and preserves its shape but not size.
- **Affine transformation** allows angular distortion of the rectangle but preserves the parallelism of lines.
- **Projective transformation** allows both angular and length distortions, thus allowing the rectangle to be transformed into an irregular quadrilateral.



Affine Transformation



Ground Control Points (GCPs)

Root Mean Square Error (RMS)

- The location of a control point on a digitized map or an image is an estimated location and can deviate from its actual location.
- A common measure of the goodness of the control points is the RMS error, which measures the deviation between the actual and estimated locations of the control points.

RMS Error

$$\sqrt{(x_{\text{act}} - x_{\text{est}})^2 + (y_{\text{act}} - y_{\text{est}})^2}$$

$$\sqrt{\left[\sum_{i=1}^{n} (x_{\text{act},\,i} - x_{\text{est},\,i})^2 + \sum_{i=1}^{n} (y_{\text{act},\,i} - y_{\text{est},\,i})^2\right] / n}$$

- Geometric transformation is an iterative process of selecting control points, estimating the transformation coefficients, and computing the RMS error. This process continues until a satisfactory transformation is obtained.

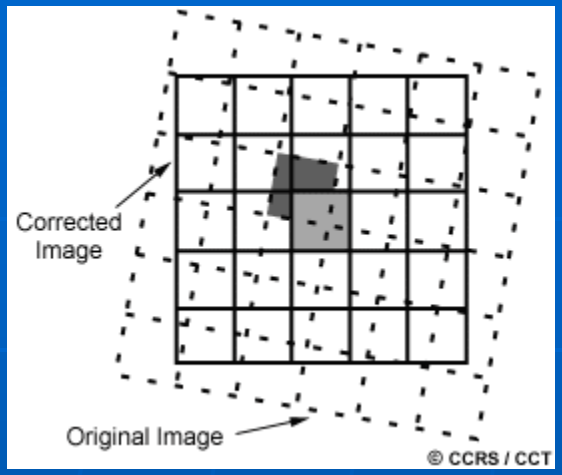
Resampling of Pixel Values

Resampling of Pixel Value

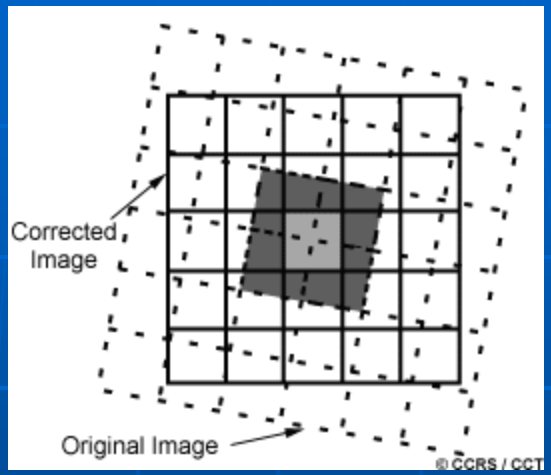
- The result of geometric transformation of the satellite image is a new image based on a projected coordinate system.
- But the new image has no pixel values.
- The pixel values must be filled through resampling.
- Resampling in this case means filling each pixel of the new image with a value or a derived value from the original image.

Resampling Methods

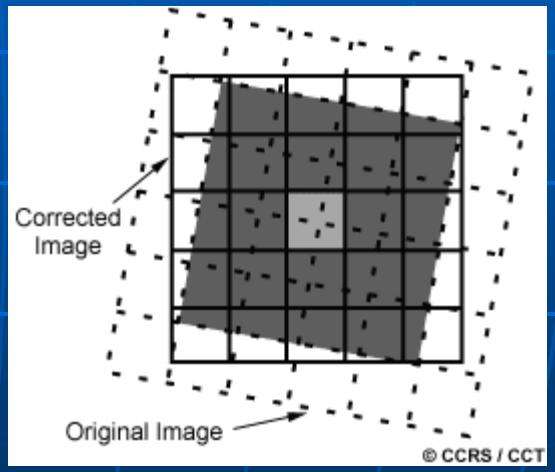
- The nearest neighborhood resampling method fills each pixel of the new image with the nearest pixel value from the original image.
- The bilinear interpolation method uses the average of the 4 nearest pixel values.
- The cubic convolution methods uses the average of the 16 nearest pixel values.



Nearest Neighborhood



Bilinear Interpolation



Cubic Convolution

Box 7.4 Computation for Bilinear Interpolation

Bilinear interpolation uses the four nearest neighbors in the original image to compute a pixel value in the new image. Pixel x in Figure 7.7 represents a pixel in the new image whose value needs to be derived from the original image. Pixel x corresponds to a location of (2.6, 2.5) in the original image. Its four nearest neighbors have the image coordinates of (2, 2), (3, 2), (2, 3), and (3, 3), and the pixel values of 10, 5, 15, and 10, respectively.

Using the bilinear interpolation method, we first perform two linear interpolations along the scan lines 2 and 3 to derive the interpolated values at a and b:

$$a = 0.6(5) + 0.4(10) = 7$$
$$b = 0.6(10) + 0.4(15) = 12$$

Then we perform the third linear interpolation between a and b to derive the interpolated value at x:

$$x = 0.5(7) + 0.5(12) = 9.5$$

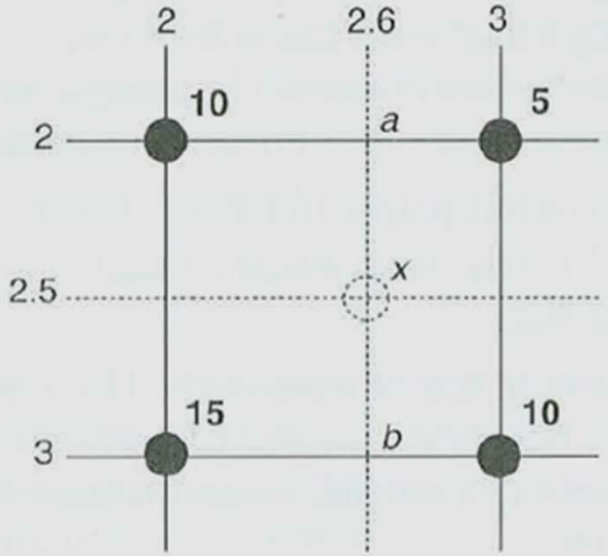
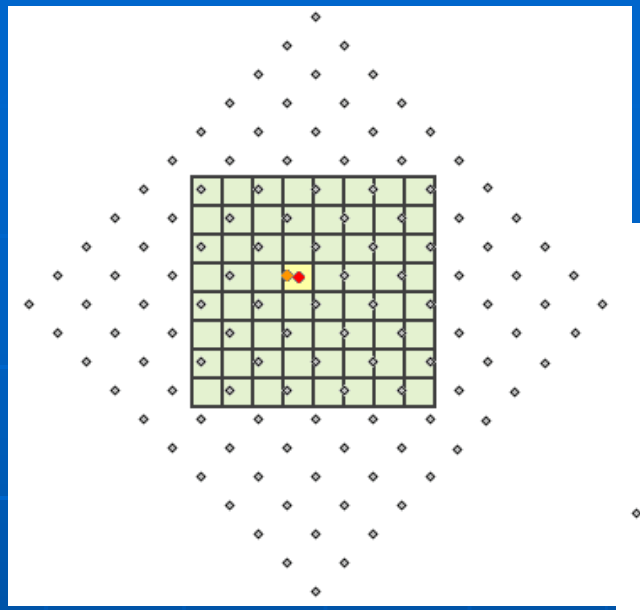
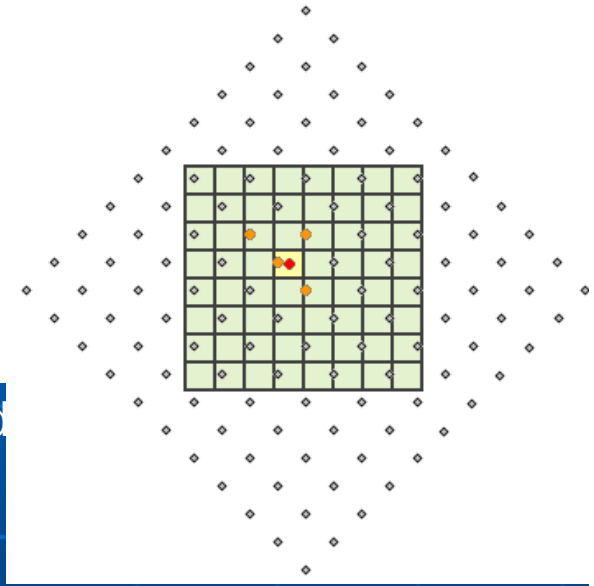


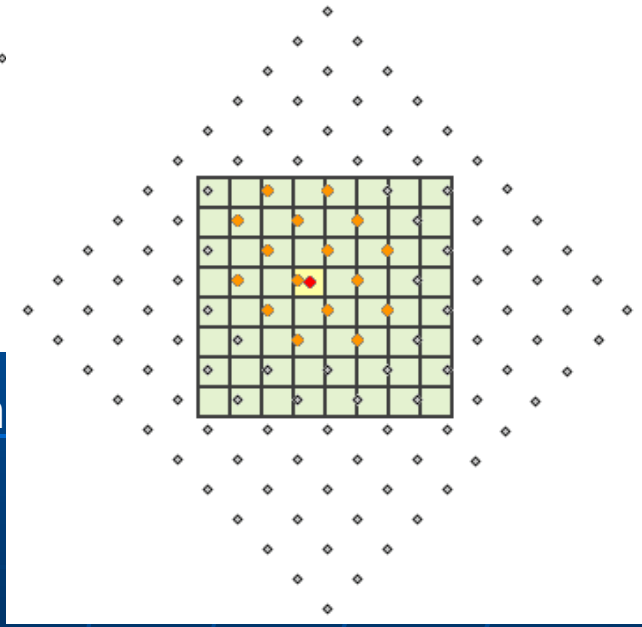
Figure 7.7
The bilinear interpolation method uses the value of the four closest pixels (black circles) in the original image to estimate the pixel value at x in the new image.



Nearest Neighborhood



Bilinear Interpolation



Cubic Convolution

- Interpolation of green pixels : the average of the upper, lower, left and right pixel values is assigned as the G value of the interpolated pixel. For example : G8 = (G3+G7+G9+G13) / 4
- Interpolation of red/blue pixels :
  - Interpolation of a red/blue pixel at a green position : the average of two adjacent pixel values in corresponding color is assigned to the interpolated pixel. For example : B7 = (B6+B8) / 2 ; R7 = (R2+R12) / 2
  - Interpolation of a red/blue pixel at a blue/red position : the average of four adjacent diagonal pixel values is assigned to the interpolated pixel. For example : R8 = (R2+R4+R12+R14) / 4 ; B12 = (B6+B8+B16+B18) / 4

